

Multiple Choice (10 marks)

1. Which of the following decimal numbers has an exact representation in binary notation?
 - (a) 0.1
 - (b) 0.2
 - (c) 0.3
 - (d) 0.5
2. Which of the following is the name of the data structure in a compiler that is responsible for managing information about variables and their attributes?
 - (a) Abstract Syntax Tree (AST)
 - (b) Attribute Grammar
 - (c) Symbol Table
 - (d) Semantic Stack
3. What is the value of this python expression: $1 + 3 \% 3$?
 - (a) 0
 - (b) 1
 - (c) 3
 - (d) 4
4. Which pillar of cybersecurity is compromised when someone logs into a system using a stolen login and password?
 - (a) Authentication
 - (b) Confidentiality
 - (c) Integrity
 - (d) Nonrepudiation
5. In Python 3, which of the following function checks in a string that all characters are in uppercase?
 - (a) `isupper()`
 - (b) `join(seq)`
 - (c) `len(string)`
 - (d) `ljust(width[, fillchar])`

True / False (10 marks)

Answer True or False

6. True or False: The output of the expression $4*1**4$ in Python is 4.
7. True or False: All bit strings whose number of zeros is a multiple of five can be described with a regular expression.
8. True or False: The Address Resolution Protocol (ARP) is used to determine the hardware address of a given IP address.
9. True or False: The page-replacement policy that minimizes page faults replaces the page whose most recent reference was the shortest time in the past.
10. True or False: Merge sort has both average-case and worst-case time complexities of $O(n \log n)$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to calculate the product of all the numbers of a given tuple.
12. Write a function to replace characters in a string.

End of Paper